Calibration Dimension of Average-Precision Ranking with Oracle Block Abstention

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Abstract

A ranking system may commit to an order between blocks while paying for an ideal resolver to order items within each block. We study average precision under this oracle block-abstention loss. A block of size at most b introduces normalized relevance interactions of degree up to $b+1$, although ordinary average precision is quadratic. We determine the exact dimension of all utility contrasts: $n + \binom{n}{2} - 2$ without abstention, and $\sum_{s=1}^{\min(b+1,n)} \binom{n}{s} - 1$ when $b \geq 2$. A fixed, nonnegative fee depending only on block position and size makes this dimension the exact convex calibration dimension. The same dimension is necessary and sufficient for a common convex surrogate with fee-dependent links across all admissible fees. We also characterize sufficient mean reports and give excess-risk bounds. For uniform pair fees, a three-item distribution fiber has identical quadratic normalized moments but incompatible optimal actions throughout a nonempty fee interval; its deterministic and randomized deficiencies are computed exactly. Higher-order parity constructions give analogous full-menu obstructions. Exact rational checks and a subset decoder validate the finite instances. The results concern ideal resolution and unrestricted conditional distributions, not trained retrieval systems.

Keywords: average precision, abstention, convex calibration dimension, structured prediction, sufficient statistics

1 Introduction

A ranker can return $1 \mid \{2,3\}$ and pay to resolve the second block. This *block abstention* defers within-block ordering without rejecting a query or item. Its average-precision (AP) value depends on the relevance law and may require richer information than choosing a complete ranking.

Our ordered-partition problem commits to block order before an ideal resolver observes binary relevance and ranks relevant items first within each block. Completed AP utility is charged a fee for each nonsingleton block. Ideal resolution isolates information geometry from resolver error, acquisition delay, and retrieval learning.

Complete AP is determined by normalized moments through degree two. Oracle resolution of size-b blocks introduces degree-$b+1$ interactions. Monomial count is not calibration dimension: shared directions cancel, and a convex lower bound requires contrasts active at a Bayes distribution. We address both conditions.

1.1 Main Results

Let $n \geq 2$, let $b \in \{1, \ldots, n\}$ be the largest allowed block, and let U_A be the oracle-completed AP utility of action A. Write $\mathcal{C}_{n,b}$ for the span of all utility contrasts $U_A - U_{A'}$. The first

result determines this space exactly:

$$D_{n,b} := \dim \mathcal{C}_{n,b} = \begin{cases} n + \binom{n}{2} - 2, & b = 1, \\ \sum_{s=1}^{\min(b+1,n)} \binom{n}{s} - 1, & b \geq 2. \end{cases} \tag{1}$$

Complete permutations preserve two independent coefficient sums. A two-item block removes one of these constraints and adds cubic directions; each larger block size adds every new interaction order permitted by the block expansion. The proof is constructive: it derives all block coefficients, builds the quadratic span from adjacent exchanges and pair merges, and then inductively generates the higher-order directions.

To activate the contrast span, a nonnegative reference-value price depending only on block position and size ties all expected priced losses under a fixed full-support reference law. The feasible-subspace calibration theorem yields

$$\mathrm{CCdim}(\ell_{c^\circ}) = D_{n,b}.$$

A squared-loss contrast-basis report attains this bound and, with fee-dependent decoders, handles every admissible schedule, making $D_{n,b}$ the minimum common convex-surrogate dimension. The equality is schedule-specific: with zero fee and $b = n$, the full block has zero loss everywhere, so a constant report suffices.

A squared mean report admits calibrated links for every fee iff its affine feature span contains $\mathcal{C}_{n,b}$. For a fixed finite loss, each report fiber need only admit one action Bayes throughout it. The robust link, not just an exact-mean decoder, bounds feature-estimation, approximate-decoding, and fee-perturbation errors separately.

For three items with a uniform unresolved-pair fee, a full one-dimensional distribution family shares normalized moments through degree two. Over a nonempty fee interval, its Bayes action switches between complete ranking and $1 \mid \{2, 3\}$; we compute deterministic/randomized minimax regret over the whole fiber. At every block order, parity laws share moments through degree b but have disjoint Bayes groups in the full menu under reference pricing. These obstruct the specified mean reports; they do not establish convex calibration dimension $D_{n,b}$ at every uniform fee.

An exact subset decoder handles moment reports with additive block fees but is exponential in n. Rational enumeration of every cap through $n = 5$, larger targeted parity checks, three mechanism-level negative controls, and exhaustive decoder comparisons certify finite proof identities, not general inference efficiency.

1.2 Scope and Organization

The theory assumes perfect within-block resolution on nonempty binary relevance vectors; it does not model resolver error, trained retrieval, or deployment. After related work, Sections 3–5 develop the loss and contrasts; Sections 6–7 give calibration and obstructions; Section 8 gives decoding and finite checks.

2 Related Work

Completion rule, link restriction, and target metric distinguish the following five literatures.

2.1 AP Optimization and Neighboring Ranking Actions

Preference-based ranking includes the boosting and gradient approaches of Freund et al. (2003); Burges et al. (2005) and the label-ranking formulation of Hüllermeier et al. (2008). Listwise methods instead address ranking lists (Cao et al., 2007; Xia et al., 2008). Direct metric approaches include boosting (Xu and Li, 2007), smooth rank approximations (Taylor et al., 2008; Qin et al., 2010), and differentiable AP objectives for image retrieval (Revaud et al., 2019; Brown et al., 2020). The historical AP ancestor, Yue et al. (2007), uses binary relevance, complete rankings, and an AP-specific structural hinge. It has neither selective blocks nor fee-dependent Bayes geometry.

Ties and partial rankings are nearby but distinct. Fagin et al. (2003) compare top-k lists, McSherry and Najork (2008) average retrieval measures over score-tie refinements, and Cheng et al. (2010) trade partial-order correctness against completeness. Wang and Scott (2020) predict ordered partitions for a multiclass discrete loss calibrated by the Weston–Watkins hinge. Their target compares a partition with one class label; ours evaluates binary relevance by AP after outcome-dependent oracle completion and adds declared fees. Outcome-independent averaging also cannot generate the higher-order terms created by relevance-aware maximization.

2.2 Ranking Consistency

Ranking consistency includes empirical and subset-ranking analyses (Clémençon et al., 2008; Cossock and Zhang, 2008); bipartite ranking and regret-transfer analyses provide another strand (Kotłowski et al., 2011; Agarwal, 2014; Menon and Williamson, 2016). Natural surrogates can nevertheless fail when conditional optima do not identify the target order (Duchi et al., 2010; Ravikumar et al., 2011). For AP, Calauzènes et al. (2012) exclude convex one-score-per-item losses used with a sorting link. More directly, Ramaswamy and Agarwal (2016, Section 5.3) study the same nonzero binary relevance space and complete-permutation actions as our $b = 1$ loss, without restricting the link, and prove $\frac{n(n-1)}{2} - 4 \leq \mathrm{CCdim}(\ell_{\mathrm{AP}}) \leq \frac{n(n+1)}{2}$. Our $b = 1$ theorem closes that interval at $\frac{n(n+1)}{2} - 2$. For $b \geq 2$, the ordered oracle-resolved blocks, declared fees, and resulting higher-order contrast and fiber geometry are additional. Top-k and ordinal calibration likewise depend on the exact target and link (Yang and Koyejo, 2020; Pedregosa et al., 2017).

2.3 Rejection, Selection, and Deferral

Classical reject-option work compares prediction error with reject cost (Chow, 1970; Herbei and Wegkamp, 2006); calibrated binary and multiclass constructions appear in Bartlett and Wegkamp (2008); Yuan and Wegkamp (2010); Ramaswamy et al. (2018), while selective classification studies coverage–risk behavior (El-Yaniv and Wiener, 2010). Rejection and deferral may invoke an expert or downstream decision maker (Cortes et al., 2016; Mozannar and Sontag, 2020; Verma and Nalisnick, 2022), and Franc et al. (2023) characterize optimal reject strategies for several objectives. These models usually reject a whole decision or instance; our committed block boundaries remain scored while only within-block order is resolved.

Ranking-specific abstention is also heterogeneous. Mao et al. (2023) analyze hypothesis-class consistency for a ranking-with-abstention loss, Gisserot-Boukhlef et al. (2024) reject whole reranking queries, and the closest recent work, Ferrara et al. (2026), selects pairs under a coverage constraint by thresholding conditional pairwise risk. None uses AP-scored ordered partitions with relevance-aware completion and additive fees. The structured abstain loss of Finocchiaro et al. (2022) instead prices erroneous and abstained coordinates through submodular costs.

2.4 Calibration, Elicitation, and Structured Surrogates

Conditional calibration links surrogate and target risk (Zhang, 2004; Bartlett et al., 2006; Tewari and Bartlett, 2007); proper composite losses expose reports and links (Reid and Williamson, 2010; Williamson et al., 2016), while elicitation studies which distributional statistics minimize expected loss (Lambert et al., 2008; Frongillo and Kash, 2015; Agarwal and Agarwal, 2015). The forecast-calibration notions of Derr et al. (2026) differ from our surrogate-to-target conditional-risk criterion; property language here describes only sufficient mean reports.

Low-rank squared surrogates give general finite-loss upper bounds (Ramaswamy et al., 2013), and least-squares analyses cover broad discrete losses, including ranking (Nowak et al., 2019). Convex calibration dimension supplies active-face lower bounds rather than rank-only conclusions (Ramaswamy and Agarwal, 2016). Structured-prediction analyses treat consistent and inconsistent surrogates (Osokin et al., 2017; Struminsky et al., 2018). Broad constructions include conditional embeddings (Ciliberto et al., 2020), smooth surrogates (Nowak-Vila et al., 2019), Fenchel–Young losses (Blondel et al., 2020), and polyhedral embeddings (Finocchiaro et al., 2024). Our squared upper bound is established machinery; the contribution is the exact oracle-AP span, a fee that activates it, and fibers showing when lower-order reports fail. Hypothesis-class cost sensitivity (Finocchiaro et al., 2025) restricts model capacity instead of the unrestricted conditional geometry studied here.

2.5 Nondecomposable Measures

AP shares with the F-measure a dependence on joint rather than coordinatewise information. Exact and Bayes-optimal F-measure analyses identify sufficient joint-distribution parameters (Dembczyński et al., 2011; Waegeman et al., 2014); generalized binary and multiclass metric frameworks optimize confusion-matrix functionals (Koyejo et al., 2014; Narasimhan et al., 2015, 2024). Multilabel F-measure surrogates (Zhang et al., 2020) and recent exact rank/dimension bounds for multilabel F_1 (Zhang, 2026) are closest in geometric spirit, but their outcomes, actions, and sufficient statistics differ. We therefore resolve one structured loss rather than claim a general theory of nondecomposable metrics.

3 The Oracle Block Decision Problem

3.1 Outcomes, Complete Rankings, and Partial Commitment

Write $[n] = \{1, \dots, n\}$. The outcome space and relevance count are

$$\mathcal{Y}_n = \{0,1\}^n \setminus \{0\}, \qquad R(y) = \sum_{i=1}^{n} y_i.$$

The model restricts attention to queries with at least one relevant item, so AP is defined without an all-zero-vector convention. Every theorem quantifies over all probability distributions on $\mathcal{Y}_n$, including dependent relevance vectors and distributions on its boundary.

For a permutation $\pi = (\pi_1, \dots, \pi_n)$, define

$$\mathrm{AP}(\pi, y) = \frac{1}{R(y)} \sum_{k=1}^{n} \frac{y_{\pi_k}}{k} \sum_{j=1}^{k} y_{\pi_j}. \tag{2}$$

The value lies in $[0,1]$ and is one when all relevant items precede all irrelevant items. Multiplying by R gives

$$Q_\pi(y) := R(y)\,\mathrm{AP}(\pi, y) = \sum_{k=1}^{n} \frac{y_{\pi_k}}{k} + \sum_{1 \le j < k \le n} \frac{y_{\pi_j} y_{\pi_k}}{k}. \tag{3}$$

We used $y_i^2 = y_i$ in this expression. Thus Q_π is its unique multilinear representation on the Boolean cube.

An action is an ordered set partition $A = (B_1, \dots, B_K)$ of $[n]$: each block is nonempty, the blocks are disjoint, and their union is $[n]$. The menu $\mathcal{A}_{n,b}$ contains exactly the actions with $|B_j| \le b$. There is no order inside a block. All items in B_j must precede all items in B_{j+1}. The complete permutations are the actions with only singleton blocks, and they are present for every b.

Let $\Pi(A)$ be the set of permutations obtained by ordering each block without changing the block order. The oracle utility is

$$U_A(y) = \max_{\pi \in \Pi(A)} \mathrm{AP}(\pi, y), \qquad Q_A(y) = R(y) U_A(y). \tag{4}$$

The maximum is taken *after* y is realized. The action is chosen before y is observed. Interchanging this maximum and an expectation would define a different decision problem. The oracle need not choose an order between equally relevant items; these orders have the same AP.

Example 1 *Take $A = \{1\} \mid \{2,3\}$. At $y = (0,1,0)$, a resolver returns $(1,2,3)$ and $U_A = 1/2$. At $y = (0,0,1)$ it returns $(1,3,2)$ and again $U_A = 1/2$. An outcome-independent uniform completion has expected AP $\frac{1}{2}(\frac{1}{2} + \frac{1}{3}) = 5/12$ at either outcome. At $y = (1,1,1)$ every completion has AP one. Hence resolution changes outcomes differently; its value is not a constant improvement attached to a block.*

3.2 Fees and Bayes Actions

An *admissible* fee schedule is a vector $c = (c_A)_{A \in \mathcal{A}_{n,b}}$ of finite nonnegative numbers, with $c_\pi = 0$ for every complete permutation. The loss is

$$\ell_c(A, y) = 1 - U_A(y) + c_A. \tag{5}$$

The fees are known to the decoder and independent of the realized outcome. They may depend on the whole action. An important subclass is additive:

$$c_A = \sum_{j=1}^{K} q(S_j, B_j), \qquad S_j = \bigcup_{i<j} B_i, \tag{6}$$

where $q(S, B)$ is known. When it depends only on $t = |S|$ and $s = |B|$, we write $q(t, s)$. The uniform pair fee is

$$c_\lambda(A) = \lambda \sum_{j=1}^{K} \binom{|B_j|}{2}, \qquad \lambda \geq 0. \tag{7}$$

It counts the within-block pairs whose order was not committed and serves as an abstract decision charge rather than a model of human labor cost.

For a conditional law $p \in \Delta(\mathcal{Y}_n)$, write

$$L_c(A; p) = \mathbb{E}_p \ell_c(A, Y), \quad L_c^*(p) = \min_{A \in \mathcal{A}_{n,b}} L_c(A; p), \quad \mathrm{Reg}_c(A; p) = L_c(A; p) - L_c^*(p).$$

The Bayes action set $\mathcal{B}_c(p)$ is the set of minimizers. Our links select one action from this finite set when possible. Ties in finite minimizations are broken by a fixed enumeration of the actions, yielding measurable links for the continuous scores used below. A randomized action is a distribution over $\mathcal{A}_{n,b}$, evaluated by its expected loss; randomization is introduced only where stated.

The conditional formulation allows a query or covariate X: apply the definitions to $p_x = \mathcal{L}(Y \mid X = x)$. We impose no parametric family for p_x and no restriction on the measurable report function $x \mapsto u(x)$. This is the standard unrestricted calibration question, not a statement about optimization in a neural or linear hypothesis class. In particular, AP averaged across queries is an expectation of the conditional AP loss, not AP computed after pooling items from different queries.

3.3 Convex Surrogates and Report Dimension

A k-dimensional convex surrogate consists of a convex report domain $\mathcal{U} \subseteq \mathbb{R}^k$ and a finite, nonnegative function $S : \mathcal{U} \times \mathcal{Y}_n \to \mathbb{R}$ that is convex in its first argument. Its conditional excess risk is

$$\delta_S(u; p) = \mathbb{E}_p S(u, Y) - \inf_{v \in \mathcal{U}} \mathbb{E}_p S(v, Y).$$

A link $\psi_c : \mathcal{U} \to \mathcal{A}_{n,b}$ is calibrated for ℓ_c if, for every p and every sequence $u_m \in \mathcal{U}$,

$$\delta_S(u_m; p) \longrightarrow 0 \quad \Longrightarrow \quad \mathrm{Reg}_c(\psi_c(u_m); p) \longrightarrow 0. \tag{8}$$

This near-minimizer formulation matters when the surrogate infimum is not attained. It also excludes a link that behaves correctly at one exact minimizer but incorrectly arbitrarily close to it. Conditional calibration underlies consistency analyses of multiclass losses (Tewari and Bartlett, 2007); continuity and approximation issues already arise for large-margin surrogates (Zhang, 2004).

The convex calibration dimension $\mathrm{CCdim}(\ell_c)$ is the smallest k for which such a surrogate and link exist. We extend the usual positive-integer convention by permitting dimension zero when a constant action is Bayes for every p; every nontrivial lower-bound application below has positive dimension. A *common* surrogate across fees is a single S, independent of c, together with a calibrated link ψ_c for every admissible c. Only the link may use the fees. The all-fee statement below is about this common report space; it does not constrain how many different optimization problems a particular training implementation might solve.

For a feature map $T : \mathcal{Y}_n \to \mathbb{R}^k$ and a positive-definite matrix W, a central example is

$$S_W(u, y) = \|u - T(y)\|_W^2 , \qquad \|v\|_W^2 = v^\top W v. \tag{9}$$

Writing $\mu(p) = \mathbb{E}_p T(Y)$, expansion of the square gives

$$\delta_{S_W}(u; p) = \|u - \mu(p)\|_W^2 . \tag{10}$$

This surrogate elicits a mean, not generally a joint probability vector. Proper composite losses analyze related probability-reporting questions (Reid and Williamson, 2010), with a multiclass development in Williamson et al. (2016). Here the essential task is to identify which *linear* property of the relevance law is sufficient for the priced ranking decision.

3.4 Normalized Interactions

For a nonempty subset $T \subseteq [n]$, let

$$y_T = \prod_{i \in T} y_i, \qquad \varphi_T(y) = \frac{y_T}{R(y)}, \qquad \mu_T(p) = \mathbb{E}_p \varphi_T(Y).$$

These are normalized relevance moments. They should not be confused with ordinary joint probabilities $\mathbb{E}y_T$. In particular, $\mathbb{E}y_i$ is not generally sufficient for AP. The identity $\sum_i \varphi_{\{i\}} = 1$ will account for one affine redundancy in the reports. For notational economy, subscripts such as 12 denote sets such as $\{1, 2\}$, never a position in a ranking.

A distribution fiber for a feature map T is the set of laws with one fixed mean $\mathbb{E}T(Y)$. If two laws in a fiber require disjoint Bayes action sets, no method whose optimal report is just this mean can be calibrated for both. This is an information obstruction, independent of numerical optimization. For nondecomposable prediction measures, the distinction between marginal and joint information is already important, for example in Bayes-optimal F-measure prediction (Waegeman et al., 2014). We will identify the exact missing interactions for oracle AP rather than assuming that all dependence must be estimated.

4 Where the Higher-Order Interactions Come From

The maximum in (4) appears to require all within-block permutations. An exchange argument removes that maximum. A finite-difference identity then reveals precisely which relevance interactions remain.

4.1 The Exact Contribution of a Resolved Block

Lemma 1 (Adjacent exchange) *Suppose items i, j occupy adjacent positions $t+1, t+2$ after a prefix P of size t. Put $a = \sum_{p \in P} y_p$. For two complete orders that differ only by swapping i, j,*

$$Q_{Pij\cdots}(y) - Q_{Pji\cdots}(y) = \frac{(y_i - y_j)(1+a)}{(t+1)(t+2)}. \tag{11}$$

Consequently, U_A is attained by placing all relevant items first in every block.

Proof Contributions before these positions agree. After the two positions, the prefix relevance count is also identical, so later contributions cancel. The remaining difference is

$$\frac{y_i(a+y_i) - y_j(a+y_j)}{t+1} + \frac{y_j(a+y_i+y_j) - y_i(a+y_i+y_j)}{t+2}.$$

Using $y_i^2 = y_i$ and $y_j^2 = y_j$ gives (11). Every adjacent irrelevant–relevant inversion inside a block can therefore be removed without decreasing AP, and strictly increases it when the indicators differ. Repeated swaps produce the stated completion while respecting every block boundary. ■

Define the shifted harmonic sum

$$h_t(r) = \sum_{k=1}^{r} \frac{1}{t+k}, \qquad h_t(0) = 0.$$

For a block B with s items, let S be its prefix set, $t = |S|$, $a = R_S(y)$, and $r = R_B(y)$. Under the completion of Lemma 1, its r relevant items occupy the first r positions of that block. Its numerator contribution is exactly

$$\sum_{k=1}^{r} \frac{a+k}{t+k} = r + (a-t)h_t(r). \tag{12}$$

In the first block, $a = t = 0$, so the contribution is just r. Later blocks are different: the denominator positions depend on the number of preceding items, whereas the precision numerators depend on the number of preceding relevant items. Their difference is what couples the block to its prefix.

Proposition 2 (Block formula) *For any $A = (B_1, \ldots, B_K)$ and any $y \in \mathcal{Y}_n$,*

$$Q_A(y) = \sum_{j=1}^{K} \Big\{ R_{B_j}(y) + \big(R_{S_j}(y) - |S_j|\big) h_{|S_j|}(R_{B_j}(y)) \Big\}. \tag{13}$$

The right side extends to the whole Boolean cube and equals zero at $y = 0$.

Proof Sum (12) over the blocks. At $y = 0$, both R_{B_j} and $h_{|S_j|}(R_{B_j})$ vanish, so the same expression is zero. ■

4.2 A Newton Expansion on a Block

The harmonic term is not linear in r, but it has a finite multilinear expansion in the indicators of the block. For $s \geq 1$, define

$$\gamma_{t,s} = (-1)^{s-1} \frac{(s-1)!t!}{(t+s)!}. \tag{14}$$

None of these coefficients is zero.

Lemma 3 (Harmonic interactions) *For a block B and $t \geq 0$,*

$$h_t(R_B(y)) = \sum_{\emptyset \neq T \subseteq B} \gamma_{t,|T|} y_T. \tag{15}$$

Proof The coefficient of a monomial on an s-element subset is the sth forward difference $\Delta^s h_t(0)$, because the function depends only on the number of active indicators. One can evaluate it without a recurrence. The integral identity

$$h_t(r) = \int_0^1 x^t \frac{1-x^r}{1-x}\, dx$$

holds by expanding the finite geometric series. Taking $s \geq 1$ forward differences at $r = 0$ gives

$$\Delta^s h_t(0) = (-1)^{s-1} \int_0^1 x^t (1-x)^{s-1}\, dx = (-1)^{s-1} \frac{t!(s-1)!}{(t+s)!}.$$

The last equality follows by repeated integration by parts. Newton's finite identity writes $h_t(r) = \sum_{s=1}^{r} \binom{r}{s} \Delta^s h_t(0)$. On a binary block, $\binom{R_B(y)}{s} = \sum_{T \subseteq B, |T|=s} y_T$, which proves the result. ∎

Combining the two formulas gives a useful coefficient-level representation:

$$Q_A(y) = \sum_{B \in A} \left[\sum_{i \in B} y_i + \sum_{\emptyset \neq T \subseteq B} \gamma_{|S|,|T|} \left(\sum_{i \in S} y_{T \cup \{i\}} - |S| y_T \right) \right], \tag{16}$$

where S is the union of blocks preceding B. Prefix and block are disjoint, so no product in this formula has repeated variables.

Theorem 4 (Interaction degree) *Let $d = \min(b+1, n)$. Every Q_A, $A \in \mathcal{A}_{n,b}$, is a multilinear polynomial of degree at most d and with zero constant coefficient. Hence every expected oracle utility is a linear function of $(\mu_T)_{1 \leq |T| \leq d}$. For $b \geq 2$ and every $3 \leq s \leq d$, an action contrast has a nonzero degree-s coefficient.*

Proof A term in (16) contains at most $|B|$ block variables and one prefix variable. This proves the upper bound and the zero constant coefficient. Dividing by R and taking expectations gives the moment representation.

For sharpness, choose any s-element set T, an item $i \in T$, and the action whose first block is $\{i\}$, whose second block is $T \setminus \{i\}$, and whose remaining blocks are singletons. The coefficient of y_T in its numerator is

$$\gamma_{1,s-1} = \frac{(-1)^{s-2}}{(s-1)s} \neq 0. \tag{17}$$

The second block is the only one capable of contributing degree s; all later singleton contributions have degree at most two. Subtracting any complete refinement removes only quadratic terms. Since $s - 1 \leq b$, this is an admissible action contrast. ■

For example, $A = 1 \mid \{2, 3\}$ has numerator

$$Q_A = y_1 + \tfrac{1}{2}(y_2 + y_3) + \tfrac{1}{2}(y_1 y_2 + y_1 y_3) + \tfrac{1}{6} y_2 y_3 - \tfrac{1}{6} y_1 y_2 y_3. \tag{18}$$

The cubic term is not a numerical artifact of maximizing over two permutations. It is the last Newton coefficient of a two-item block multiplied by the prefix relevance. At the all-relevant outcome, the quadratic and cubic block corrections cancel appropriately; at other outcomes they change the value of resolution.

4.3 Why Random Ties Do Not Have the Same Effect

Proposition 5 (Outcome-independent completion) *Suppose an action A is completed using fixed weights $w_{A,\pi} \geq 0$, $\sum_{\pi \in \Pi(A)} w_{A,\pi} = 1$, independent of y. Its expected-completion utility*

$$\overline{U}_A(y) = \sum_{\pi \in \Pi(A)} w_{A,\pi} \, \mathrm{AP}(\pi, y)$$

has a quadratic numerator. If the full action menu contains the complete permutations and all charges are nonnegative, a randomized-completion action cannot have strictly smaller conditional loss than every complete permutation.

Proof The numerator is a fixed convex combination of the quadratic polynomials (3). For any p, its uncharged expected loss is the same convex combination of the complete-ranking losses. It is at least their minimum, and adding a nonnegative charge cannot reduce it. ■

This proposition is a useful negative control. It explains why the tie-aware averaging of McSherry and Najork (2008) cannot be substituted for the oracle maximum while preserving our conclusions. It also locates the mechanism: high-order information is introduced by outcome-dependent selective resolution, not by merely writing a ranking as blocks.

5 The Exact Space of Decision-Relevant Contrasts

An upper bound on polynomial degree does not determine the number of reports a calibrated surrogate needs. Some coefficients may always occur together; others may add the same value to every action. We therefore work with utility differences.

Fix a complete permutation A_0. Define

$$\mathcal{C}_{n,b} = \mathrm{span}\{U_A - U_{A_0} : A \in \mathcal{A}_{n,b}\} \subseteq \mathbb{R}^{\mathcal{Y}_n}.$$

This definition does not depend on the choice of A_0. Multiplication by $R(y) > 0$ is an invertible diagonal transformation on functions on $\mathcal{Y}_n$. We may therefore calculate the dimension in numerator polynomials and then divide by R.

5.1 Independence and the Complete-Ranking Subspace

For $1 \le d \le n$, let

$$\mathcal{M}_d = \operatorname{span}\{y_T : 1 \le |T| \le d\}, \qquad m_d = \sum_{s=1}^{d} \binom{n}{s}.$$

We identify these polynomials with their restrictions to $\mathcal{Y}_n$.

Lemma 6 (No hidden polynomial relation) *The m_d functions y_T, and likewise the m_d functions φ_T, are linearly independent on $\mathcal{Y}_n$. In the normalized span, $1 = \sum_i \varphi_{\{i\}}$ is the unique representation of the constant function one.*

Proof If a zero-constant multilinear polynomial vanishes on all nonzero Boolean vectors, it also vanishes at zero. A multilinear polynomial vanishing on the entire Boolean cube has every coefficient zero: evaluate first on singleton supports, then on supports of size two, and continue by induction on support size. This proves independence of y_T. Multiplication by R proves independence of φ_T. The asserted constant representation follows from $\sum_i y_i = R$ and is unique by that independence. ■

Let $H_n = \sum_{k=1}^{n} 1/k$. In every complete-ranking numerator, the sum of singleton coefficients is H_n, while the sum of pair coefficients is

$$\sum_{k=2}^{n} \frac{k-1}{k} = n - H_n.$$

Thus a complete-ranking contrast has zero coefficient sum separately at degree one and degree two. The next lemma shows that these are the only restrictions.

Lemma 7 (Complete-ranking contrasts) *For $n \ge 2$, the numerator contrasts of complete rankings are exactly*

$$\left\{ \sum_i a_i y_i + \sum_{i<j} a_{ij} y_i y_j : \sum_i a_i = 0, \quad \sum_{i<j} a_{ij} = 0 \right\}. \tag{19}$$

Their dimension is $n + \binom{n}{2} - 2$.

Proof The coefficient-sum identities give containment in (19). For the reverse containment, swap the first two items i, j. Equation (11) gives a nonzero multiple of $y_i - y_j$. These span every singleton coefficient vector with zero sum.

For three distinct items k, i, j, put k first and swap i, j immediately after it. The resulting contrast is a nonzero multiple of

$$(y_i - y_j)(1 + y_k).$$

Subtract a suitable multiple of an already available singleton difference. This leaves $y_i y_k - y_j y_k$. Pair monomials are vertices of a graph in which two vertices are adjacent when the pairs share an item. This graph is connected for $n \geq 3$: disjoint pairs can be joined through a pair using one item from each, and intersecting pairs are adjacent. Edge differences therefore span every pair coefficient vector with zero sum. For $n = 2$, there is one pair monomial and its zero-sum subspace has dimension zero, so no three-item argument is needed.

The singleton and pair coefficient spaces are independent by Lemma 6. Their dimensions are $n - 1$ and $\binom{n}{2} - 1$, respectively, including the zero-dimensional pair case. This proves equality and the dimension. ■

The complete-ranking space is smaller than the whole quadratic normalized-moment space by two linear coefficient constraints. One constraint expresses equality at the all-relevant outcome. The other will disappear when the first block is allowed to contain two items.

5.2 Two-Item Blocks Connect the Coefficient Levels

Let

$$\mathcal{W}_d = \{q \in \mathcal{M}_d : q(\mathbf{1}) = 0\}.$$

It has dimension $m_d - 1$. All oracle numerator contrasts lie in $\mathcal{W}_d$, since $Q_A(\mathbf{1}) = n$ for every A.

Lemma 8 (The quadratic bridge) *For every $n \geq 2$ and $b \geq 2$, oracle numerator contrasts span all of $\mathcal{W}_2$.*

Proof Place a block $\{i, j\}$ first and all remaining items in fixed singleton order. Compare it with its complete refinement having i before j. The contribution of the first block is $y_i + y_j$; that of the first two singleton blocks is $y_i + y_j/2 + y_i y_j/2$. Later contributions are unchanged because their prefix set and prefix relevance count agree. The difference is

$$\tfrac{1}{2}(y_j - y_i y_j).$$

Using the reverse refinement also gives $\frac{1}{2}(y_i - y_i y_j)$.

Consider the incidence graph whose vertices are all singleton and pair monomials, with an edge from y_i to $y_i y_j$ whenever $i \in \{i, j\}$. This graph is connected: each pair joins its two singleton endpoints, and the complete graph on the singleton labels is connected. Its edge differences span the hyperplane of coefficient vectors whose total sum is zero. That hyperplane is exactly $\mathcal{W}_2$. ■

The bridge has a direct interpretation. A complete permutation can exchange importance between singleton coefficients, or between pair coefficients. Resolving its first pair permits an exchange between a singleton coefficient and a pair coefficient. This supplies the previously missing quadratic direction before any higher-order terms are considered.

5.3 Every Higher-Order Direction Is Present

Theorem 9 (Exact contrast dimension) *For $n \geq 2$ and $1 \leq b \leq n$, the utility contrast dimension is* (1)*. More precisely, when $b \geq 2$ and $d = \min(b+1, n)$, multiplication by R maps $\mathcal{C}_{n,b}$ onto $\mathcal{W}_d$.*

Proof The case $b = 1$ is Lemma 7. Assume $b \geq 2$. Theorem 4 and equality of every $Q_A(\mathbf{1})$ give the upper containment in $\mathcal{W}_d$. Lemma 8 proves that $\mathcal{W}_2$ is already spanned.

Proceed by induction on $s = 3, \ldots, d$. Suppose all of $\mathcal{W}_{s-1}$ is spanned. For an arbitrary s-element set T, use the action in the sharpness proof of Theorem 4 and subtract a complete refinement. Call its numerator contrast q_T. It has exactly one degree-s monomial, with coefficient $a_s = \gamma_{1,s-1} \neq 0$, and has no terms of higher degree. Fix any singleton monomial y_j. The polynomial

$$q_T - a_s(y_T - y_j)$$

has degree at most $s-1$, zero constant coefficient, and value zero at $\mathbf{1}$. It belongs to $\mathcal{W}_{s-1}$ and is therefore already in the contrast span. Subtraction and division by a_s show that $y_T - y_j$ is in that span.

Doing this for every T of size s adjoins every new degree-s coefficient direction subject only to total coefficient sum zero. Explicitly, a polynomial in $\mathcal{W}_s$ can have each of its degree-s terms removed with a multiple of $y_T - y_j$, leaving a polynomial in $\mathcal{W}_{s-1}$. Hence all of $\mathcal{W}_s$ is spanned. Induction reaches d, proving equality. Lemma 6 gives dimension $m_d - 1$, and division by R preserves it. ■

Corollary 10 (Explicit contrast coordinates) *The following features form a basis of $\mathcal{C}_{n,b}$:*

$$b = 1: \quad \varphi_i - \varphi_1 \quad (i \neq 1), \qquad \varphi_T - \varphi_{12} \quad (|T| = 2,\ T \neq \{1,2\}); \tag{20}$$

$$b \geq 2: \quad \varphi_T - \varphi_1 \quad (1 \leq |T| \leq d,\ T \neq \{1\}). \tag{21}$$

In particular, no nonzero constant function belongs to $\mathcal{C}_{n,b}$.

Proof The coefficient restrictions in Lemma 7 give (20); the single coefficient-sum restriction in $\mathcal{W}_d$ gives (21). Independence follows from Lemma 6. Every contrast vanishes at $\mathbf{1}$, whereas a nonzero constant does not. ■

These coordinates are executable rather than just existential. Expand $Q_A - Q_{A_0}$ using (16). For $b \geq 2$, retain the coefficient of every non-anchor monomial; the anchor coefficient is minus their sum. For $b = 1$, perform that operation separately at the singleton and pair levels. The resulting coefficients express $U_A - U_{A_0}$ in the displayed basis. There is no need to run a generic matrix factorization to build the surrogate.

For fixed $b \geq 2$, $D_{n,b}$ grows polynomially in n with degree $b+1$. It reaches $2^n - 2$, the affine dimension of the full outcome simplex, exactly when $b \geq \max\{2, n-1\}$. Thus $D_{2,1} = 1$ is not saturated, whereas $D_{2,2} = 2$ and $D_{3,2} = 6$ are. Saturation does not mean that every fee makes full distribution information necessary; it means that every possible

decision-relevant contrast is available in the menu. Section 6 identifies a fee at which all those directions are locally relevant at once.

It is also useful to distinguish the minimal contrast report from the full list of normalized moments. For three items, ordinary AP has contrast dimension four, while all normalized moments through degree two have affine dimension five. Our three-item counterexample in Section 7 fixes the larger, five-dimensional information set. Its failure is therefore stronger than a failure of a particular minimal ordinary-AP report.

6 Calibration, Pricing, and Sufficient Reports

We first construct a fixed fee at which the contrast dimension is an exact convex lower bound. We then separate two questions: when a mean report works for every fee, and when it works for one fixed loss. The generic surrogate and fiber arguments are useful tools, but the pricing and contrast calculations are specific to oracle AP.

6.1 A Fixed Reference-Value Fee

Define the full-support distribution

$$\nu_n(y) = \frac{R(y)}{n2^{n-1}}, \qquad y \in \mathcal{Y}_n. \tag{22}$$

It is normalized because the sum of $R(y)$ over the Boolean cube is $n2^{n-1}$. Its normalized moments and the expected utility of every complete permutation are

$$\mathbb{E}_{\nu_n}\varphi_T = \frac{2^{1-|T|}}{n}, \qquad \mathbb{E}_{\nu_n}\,\mathrm{AP}(\pi, Y) = \kappa_n := \frac{n + H_n}{2n}. \tag{23}$$

The distribution ν_n is a declared pricing reference. It is *not* an assumption about the true conditional law and is not estimated from the examples in Section 7.

For $t \geq 0$, $s \geq 1$, and $t + s \leq n$, let Z_s have the binomial distribution with parameters $s, 1/2$. Define

$$c_n^\circ(t, s) = \frac{s - h_t(s) - 2t\mathbb{E}h_t(Z_s)}{2n}, \qquad c^\circ(A) = \sum_{j=1}^{K} c_n^\circ(|S_j|, |B_j|). \tag{24}$$

This schedule is computable from n, t, s alone. It assigns the same price to blocks of the same size in the same position, regardless of their item identities.

Theorem 11 (Reference pricing) *The fee* (24) *is nonnegative, is zero on singleton blocks, and satisfies*

$$c^\circ(A) = \mathbb{E}_{\nu_n}U_A(Y) - \kappa_n, \qquad L_{c^\circ}(A; \nu_n) = 1 - \kappa_n \quad \textit{for every } A. \tag{25}$$

For two-item blocks,

$$c_n^\circ(t, 2) = \frac{1}{4n(t+1)}. \tag{26}$$

Every block with $s \geq 2$ has strictly positive reference fee.

Proof Let Y be uniform on the full Boolean cube, including zero, and use the zero extension of numerator polynomials. For any such numerator Q,

$$\mathbb{E}_{\nu_n} \frac{Q(Y)}{R(Y)} = \frac{2}{n} \mathbb{E}_{\mathrm{unif}} Q(Y),$$

where the left expectation is only on $\mathcal{Y}_n$. Under the uniform law, the prefix relevance count a and block count r are independent, $\mathbb{E}a = t/2$, and $r \sim Z_s$. Formula (12) gives the expected oracle block numerator

$$\frac{s}{2} - \frac{t}{2} \mathbb{E} h_t(Z_s).$$

For any fixed complete ordering of this block, the contribution at its kth position has expectation

$$\frac{t/4 + 1/2 + (k-1)/4}{t+k} = \frac{1}{4} + \frac{1}{4(t+k)}.$$

Its block expectation is $s/4 + h_t(s)/4$. Subtracting and multiplying by $2/n$ yields (24). Summing over blocks compares U_A with any complete refinement. Equation (3) gives the same complete-ranking reference expectation κ_n for every permutation, proving (25).

Each block contribution of oracle completion is at least that of a fixed within-block ordering by the adjacent-exchange argument. Hence its reference expectation difference, and thus its fee, is nonnegative. A singleton has no choice to resolve, so its fee is zero. If $s \geq 2$, choose an outcome with an irrelevant item immediately before a relevant item in the fixed block ordering. The exchange gain is strictly positive and that outcome has positive reference mass, so the expected gain is strictly positive. Finally, substituting the three binomial probabilities for Z_2 into (24) simplifies to (26). ■

Only position–size pairs (t, s) need prices: each block pays its expected improvement over a fixed completion under ν_n. Equation (25) and $U_A \leq 1$ give $0 \leq c^\circ(A) \leq 1 - \kappa_n < 1/2$. Pair prices decrease with position; replacing this schedule by a uniform pair charge would require a separate convex-dimension proof.

6.2 From the Bayes Tie to the Convex Lower Bound

We state the imported geometric tool in the notation needed here. For a finite loss with N outcomes, the Bayes region of action a is

$$\mathcal{Q}_a = \{p \in \Delta_N : (\ell_a - \ell_j)^\top p \leq 0 \text{ for every } j\}.$$

At $p \in \mathcal{Q}_a$, let $\mu_{\mathcal{Q}_a}(p)$ be the dimension of its two-sided feasible direction space: directions along which a sufficiently short segment centered at p stays in $\mathcal{Q}_a$. Theorem 16 of Ramaswamy and Agarwal (2016) gives

$$\mathrm{CCdim}(\ell) \geq |\operatorname{supp}(p)| - \mu_{\mathcal{Q}_a}(p) - 1. \tag{27}$$

This bound is not an assertion that loss-matrix rank is always a convex calibration lower bound. Its hypotheses match the present finite, nonnegative loss. For each fixed p, finitely many non-Bayes actions have a positive minimum target regret; hence a wrong-action

sequence with surrogate excess tending to zero violates our definition exactly when the source's positive calibration gap fails. We must still calculate the feasible direction dimension at an actual Bayes distribution.

Theorem 12 (Convex dimension) *For the fixed reference fee c°,*

$$\mathrm{CCdim}(\ell_{c^\circ}) = D_{n,b}. \tag{28}$$

Moreover, the smallest dimension of a common convex surrogate calibrated, with fee-dependent links, for every admissible fee schedule is $D_{n,b}$.

Proof Put $N = 2^n - 1$ and write $d_A = U_A - U_{A_0}$ as a vector in $\mathbb{R}^N$. At reference pricing, the loss differences are negatives of the centered contrasts

$$d'_A = d_A - (\mathbb{E}_{\nu_n} d_A)\mathbf{1}.$$

The centering map is injective on $\mathcal{C}_{n,b}$: its kernel consists of constant functions, and Corollary 10 says that the contrast space contains no nonzero constant. Thus the d'_A span a $D_{n,b}$-dimensional space. They all have zero inner product with ν_n, whereas $\mathbf{1}^\top \nu_n = 1$, so $\mathbf{1}$ is independent of their span.

By Theorem 11, every action is Bayes at ν_n. At this full-support law, nonnegativity constraints of the simplex are locally inactive. A direction is feasible in both signs within the Bayes region of A_0 exactly when it preserves normalization and all the active loss equalities. Its space is therefore

$$\{v \in \mathbb{R}^N : \mathbf{1}^\top v = 0,\ (d'_A)^\top v = 0 \text{ for every } A\},$$

of dimension $N - 1 - D_{n,b}$. Applying (27) gives the lower bound $D_{n,b}$.

For the upper bound, let $T_1, \ldots, T_D$ be the explicit basis in Corollary 10, with $D = D_{n,b}$. Write $d_A = \beta_A^\top T$ and use $S(u, y) = \|u - T(y)\|^2$ on $\mathbb{R}^D$. Decode by maximizing $\beta_A^\top u - c_A$. Equation (10) implies that vanishing surrogate regret makes all estimated utility contrasts converge to their true values. Since the action set is finite, the decoded excess loss tends to zero; the quantitative argument appears in Theorem 15. This works for every admissible fixed c, with the same S. It establishes both the fixed-fee upper bound and the common-surrogate upper bound. The latter cannot be smaller than the fixed-fee lower bound already proved. ■

For $b = 1$, the zero-fee loss has the same nonzero binary outcome space and complete-permutation actions as the MAP loss of Ramaswamy and Agarwal (2016, Section 5.3). Their link-unrestricted bounds are $\frac{n(n-1)}{2} - 4 \le \mathrm{CCdim}(\ell_{\mathrm{AP}}) \le \frac{n(n+1)}{2}$; Equation (28) sharpens them to the exact value $\frac{n(n+1)}{2} - 2$. The $b \ge 2$ theorem adds the oracle-block action space, the reference fee that makes its contrasts Bayes-active, and higher-order interactions. The dimension reaches $2^n - 2$ exactly when $b \ge \max\{2, n-1\}$, so $(n, b) = (2, 1)$ is not a saturated case.

Proposition 13 (A fixed-fee collapse) *When $b = n$ and $c_A = 0$ for every action, the loss has convex calibration dimension zero.*

Proof The one-block action ($[n]$) lets the oracle put all relevant items first. Its utility is one for every $y \in \mathcal{Y}_n$, so its loss is identically zero and it is universally Bayes. A constant decoder suffices. ■

This example prevents a misleading interpretation of Theorem 12. Adding more actions can increase the available contrast dimension while a particular price makes those contrasts irrelevant to Bayes decisions. The reference-price theorem concerns a specified loss, not merely an action menu.

6.3 Which Mean Reports Work for Every Fee?

For a feature map $T : \mathcal{Y}_n \to \mathbb{R}^k$, write

$$\mathcal{V}_T = \operatorname{span}\{\mathbf{1}, T_1, \ldots, T_k\}.$$

The added constant permits an affine representation of utility contrasts.

Theorem 14 (All-fee mean-report criterion) *The squared surrogate* (9)*, on all of* $\mathbb{R}^k$*, admits calibrated fee-dependent links for every admissible fee if and only if*

$$\mathcal{C}_{n,b} \subseteq \mathcal{V}_T. \tag{29}$$

Necessity already holds if calibration is required only at the fixed reference schedule c°*.*

Proof If (29) holds, write $U_A - U_{A_0} = a_A + \beta_A^\top T$ and decode by maximizing $a_A + \beta_A^\top u - c_A$. The same argument as the upper bound in Theorem 12 proves calibration.

Conversely, suppose some contrast is outside $\mathcal{V}_T$. Finite-dimensional linear algebra gives a vector $v \in \mathbb{R}^N$ orthogonal to $\mathbf{1}$ and every feature row but not to that contrast. In particular, $\mathbf{1}^\top v = 0$ and $Tv = 0$. The full support of ν_n lets us choose $\epsilon > 0$ such that

$$p^+ = \nu_n + \epsilon v, \qquad p^- = \nu_n - \epsilon v$$

are probability distributions. They have the same feature mean. At reference pricing, all actions tie at ν_n, while their priced-utility slopes along v are not all equal. The Bayes actions at p^+ maximize these slopes and those at p^- minimize them. These sets are disjoint. The squared surrogate has the same unique minimizer at both laws, so a link of that minimizer cannot be Bayes at both. The constant sequence at this minimizer violates (8) for at least one law. ■

The criterion concerns the span of a chosen mean report, not whether a learning algorithm happens to fit that mean accurately on a particular sample. It also differs from strict properness for the full joint distribution. A proper joint-distribution surrogate is an obvious sufficient baseline, but Theorem 9 can substantially reduce the required report when b is fixed.

Since $\dim \mathcal{C}_{n,b} = D_{n,b}$ and $\mathcal{C}_{n,b}$ contains no nonzero constant, (29) forces $k \geq D_{n,b}$ after redundant coordinates are removed; the contrast features attain equality.

Conditional mean embeddings provide a general route to structured prediction (Ciliberto et al., 2020). Fenchel–Young losses place squared and other convex constructions in a broader variational framework (Blondel et al., 2020). We use the elementary square because its exact excess risk makes the information requirement transparent.

6.4 Regret Transfer, Approximate Decoding, and Price Error

The link can be approximate without losing a quantitative guarantee. Suppose

$$U_A - U_{A_0} = a_A + \beta_A^\top T,$$

and define

$$B_W = \max_{A,A'} \|\beta_A - \beta_{A'}\|_{W^{-1}}.$$

The constant is finite for a fixed problem. It depends on the chosen feature coordinates and metric, as a norm-based calibration bound should.

Theorem 15 (Excess-risk transfer) *Let $\widetilde{c}$ satisfy $\max_A |\widetilde{c}_A - c_A| \le \tau$. Suppose $\widehat{A}$ is within $\eta \ge 0$ of the maximum estimated score:*

$$a_{\widehat{A}} + \beta_{\widehat{A}}^\top u - \widetilde{c}_{\widehat{A}} \ge \max_A (a_A + \beta_A^\top u - \widetilde{c}_A) - \eta.$$

Then, for every p,

$$\mathrm{Reg}_c(\widehat{A}; p) \le B_W \sqrt{\delta_{S_W}(u; p)} + \eta + 2\tau. \tag{30}$$

For query-dependent reports, averaging gives the same bound with the square root of expected surrogate excess risk and the expected decoder and fee-error terms.

Proof Let A^* be Bayes and $\mu = \mathbb{E}_p T(Y)$. The true score difference between A^* and $\widehat{A}$ equals their loss regret. Insert and subtract their estimated scores. Approximate maximization bounds the estimated difference by η, and the two fee errors contribute at most 2τ. The remaining term is

$$(\beta_{A^*} - \beta_{\widehat{A}})^\top (\mu - u) \le \left\|\beta_{A^*} - \beta_{\widehat{A}}\right\|_{W^{-1}} \|\mu - u\|_W.$$

Equation (10) proves (30). Apply the same conditional inequality at each query and use Jensen's inequality for the square root to obtain the population statement, assuming the displayed expectations are finite. ■

For the explicit contrast coordinates, the transfer constant has a bound that does not require enumerating the action menu. This is useful even though the bound is not generally tight.

Corollary 16 (An explicit transfer constant) *Use the basis in Corollary 10 and the unweighted square, $W = I$. Then*

$$B_I \le \begin{cases} 2n, & b = 1, \\ 2n\left(1 + \dfrac{2(2^b - 1)}{b}\right), & b \ge 2. \end{cases} \tag{31}$$

Together with (30)*, this gives an explicit bound for every admissible fee, with the same feature coordinates.*

Proof Let $\|Q\|_{\mathrm{coef},1}$ denote the sum of the absolute values of the multilinear coefficients of a numerator polynomial. For a complete permutation all coefficients are nonnegative and their total is $Q_\pi(\mathbf{1}) = n$. Thus the coefficient norm of a difference of two complete numerators is at most $2n$.

For a general block action, the $\sum_{i\in B} y_i$ terms in (16) contribute at most n in total. A nonempty subblock T of size r at prefix length t contributes at most $2t|\gamma_{t,r}|$ to the coefficient norm: there are t prefix monomials and one degree-r monomial with coefficient magnitude $t|\gamma_{t,r}|$. For $t = 0$ this is zero. For $t \geq 1$,

$$t|\gamma_{t,r}| = \frac{t}{r\binom{t+r}{r}} \leq 1.$$

Therefore a block of size s contributes at most $2(2^s - 1)$ beyond its singleton terms. The ratio $(2^s - 1)/s$ is nondecreasing for positive integers s: cross multiplication for consecutive integers reduces to $(s - 1)2^s + 1 \geq 0$. Since the block sizes sum to n and each is at most b,

$$\|Q_A\|_{\mathrm{coef},1} \leq n + 2\sum_{B\in A}(2^{|B|} - 1) \leq n\left(1 + \frac{2(2^b - 1)}{b}\right).$$

In the displayed contrast bases, a coefficient vector for $U_A - U_{A'}$ is obtained by deleting the anchor coefficient or coefficients from $Q_A - Q_{A'}$. Its Euclidean norm is no larger than the full coefficient ℓ_1 norm. The triangle inequality and the preceding bounds give (31). ■

For a fixed cap, the displayed coefficient bound is linear in n, while the number of report coordinates can have degree $b + 1$ in n. This does not remove a statistical cost of estimating those coordinates: the surrogate excess risk is the sum of their squared estimation errors in this metric. Rescaling the coordinates changes both the coefficient constant and that surrogate risk. The bound should therefore be read with its stated feature normalization, rather than as an invariant comparison between training objectives.

Square-root transfer is familiar in ranking regret analysis, for instance through strongly proper losses for bipartite ranking (Agarwal, 2014). Our role for the bound is to make the AP block feature map and fee perturbation explicit. The theorem is not a sample-complexity result: estimating the mean in a restricted hypothesis class introduces approximation and statistical errors not modeled here.

The exponent cannot be increased uniformly even in the smallest complete-ranking problem. For $n = 2$, take the sufficient feature $T(y) = \frac{1}{2}(1 + \varphi_1(y) - \varphi_2(y))$ on all of $\mathcal{Y}_2$; the complete-order utility difference is $T - 1/2$. Put all probability on $(1, 0)$ and $(0, 1)$, where $T = y_1$, and let $p((1, 0)) = 1/2 + t$ for $0 < t < 1/2$. The scalar report $u = 1/2 - t$ has squared excess risk $4t^2$ and selects the wrong complete order under the Bayes threshold link. Its AP loss regret is t, since the two orders differ in utility by $1/2$ on each singleton outcome. Thus no bound $C\delta_S^\alpha$ with fixed C and $\alpha > 1/2$ can hold over these reports as t tends to zero. This is an exponent statement for squared mean estimation, not a sharp claim about the dimension dependence of B_W.

6.5 The Exact Criterion for One Fixed Fee

Span sufficiency is deliberately uniform over prices. A fixed loss can permit a coarser mean, because some actions may be irrelevant. We now give a criterion that captures this distinction for any finite target loss, not just AP. Its connection to property elicitation is conceptual rather than an additional AP-specific dimension theorem.

Let $\ell_a \in \mathbb{R}^N$ be the loss vector of action a, let T be a fixed feature map, and put

$$P = \operatorname{conv}\{T(y) : y \in \mathcal{Y}\}, \qquad F_\mu = \{p \in \Delta_N : Tp = \mu\}.$$

For $\mu \in P$, define

$$g_a(\mu) = \max_{p \in F_\mu} \big(\ell_a^\top p - \min_j \ell_j^\top p\big), \qquad \rho(\mu) = \min_a g_a(\mu). \tag{32}$$

The scalar $\rho(\mu)$ is the best worst-case regret of one deterministic action when only that feature mean is known.

Theorem 17 (Fixed-loss fibers) *The surrogate* (9) *with $W = I$ admits a calibrated link for a finite loss ℓ if and only if*

$$\bigcap_{p \in F_\mu} \mathcal{B}(p) \neq \emptyset \quad \textit{for every } \mu \in P. \tag{33}$$

Equivalently, $\rho(\mu) = 0$ for every feasible mean. Moreover, each g_a is Lipschitz on P. For some finite constant K depending on T and ℓ, the link that projects u onto P and minimizes g_a at the projected mean satisfies

$$\operatorname{Reg}(\psi(u); p) \le \sup_{\mu \in P} \rho(\mu) + K\sqrt{\delta_S(u; p)}. \tag{34}$$

At an exact mean, the smallest deterministic worst-fiber regret is precisely $\rho(\mu)$.

Proof If the intersection is empty, no action is Bayes for every law in that fiber. The unique squared-risk minimizer is μ for all of them. Any link at μ therefore fails calibration for at least one law, using a constant report sequence. Nonnegative regrets show directly that (33) is equivalent to $\rho(\mu) = 0$.

We prove sufficiency with a continuity bound, rather than assuming that exact-mean optimality automatically controls nearby reports. For each pair of actions put

$$v_{aj}(\mu) = \max\{(\ell_a - \ell_j)^\top p : p \ge 0,\ \mathbf{1}^\top p = 1,\ Tp = \mu\}.$$

Then $g_a = \max_j v_{aj}$. Use coordinates intrinsic to the affine hull of P, removing redundant rows from the combined normalization and moment constraints. Their matrix M has full row rank and their right side $b(\mu)$ is affine in μ. Finite linear-programming duality gives

$$v_{aj}(\mu) = \min\{b(\mu)^\top z : M^\top z \ge \ell_a - \ell_j\}.$$

The dual feasible polyhedron is nonempty: the normalization coordinate can dominate every entry of the finite loss difference. Its lineality is zero because M has full row rank. For every

feasible μ, the primal optimum is finite and attained, and a dual optimal face has a vertex. There are finitely many dual vertices. Consequently v_{aj} is the minimum of finitely many affine functions on P, so it is Lipschitz there. Taking a finite maximum over j preserves Lipschitz continuity. Choose a common finite constant K for all g_a.

Let $\widehat{\mu}$ be the Euclidean projection of u onto the compact convex set P and choose, with a fixed tie rule, an action a minimizing $g_a(\widehat{\mu})$. The functions are continuous and the action set is finite, so this link is measurable. For the true $\mu = Tp$,

$$\mathrm{Reg}(a;p) \le g_a(\mu) \le g_a(\widehat{\mu}) + K\,\|\mu - \widehat{\mu}\| \le \sup_{\mu' \in P} \rho(\mu') + K\,\|\mu - u\|\,.$$

The last inequality uses nonexpansiveness of projection and $\mu \in P$. The square identity proves (34). If ρ is identically zero, this is calibration. At an exact mean, definition (32) proves the claimed minimax value, and a minimizing action exists because the action set is finite. ■

A randomized action has zero worst-fiber regret only if every action in its support is Bayes for every distribution in the fiber: every individual regret is nonnegative. Thus randomization cannot repair a zero-floor failure of (33), although it can reduce a positive minimax floor. Section 7 computes that reduction exactly.

The theorem is a finite conditional characterization; it does not claim that constructing all fiberwise linear programs is computationally efficient. Nor does it replace the general study of inconsistent structured surrogates in Struminsky et al. (2018). It identifies the precise information obstacle for a specified mean map and gives a robust link when the obstacle is absent.

7 Distribution Fibers and Unavoidable Decision Error

The dimension theorem gives an exact answer at reference pricing, but it does not by itself show how a familiar low-order report fails. We give two complementary constructions. The first scales with block size and concerns the full reference-priced menu. The second uses a uniform pair fee, fixes an entire quadratic-moment fiber, and gives its exact minimax deficiency.

7.1 A Parity Obstruction at Every Interaction Order

Fix $b \ge 2$ and put $n = b + 1$. Define $Z = n2^{n-2}$ and two laws on $\mathcal{Y}_n$ by

$$p^+(y) = \frac{R(y)}{Z}\mathbf{1}\{n - R(y) \text{ is even}\}, \qquad p^-(y) = \frac{R(y)}{Z}\mathbf{1}\{n - R(y) \text{ is odd}\}. \tag{35}$$

The relevance weighting in these laws is important. Ordinary uniform parity laws match unnormalized low-order moments, whereas AP requires moments divided by R.

Lemma 18 (Matching normalized moments) *The laws* (35) *are probability distributions, their midpoint is ν_n, and*

$$\mathbb{E}_{p^+}\varphi_T = \mathbb{E}_{p^-}\varphi_T = \frac{2^{1-|T|}}{n} \quad (1 \le |T| \le b), \qquad \mathbb{E}_{p^+}\varphi_{[n]} - \mathbb{E}_{p^-}\varphi_{[n]} = \frac{1}{Z}. \tag{36}$$

Proof On the full cube, the signed sum $\sum_y (-1)^{n-R(y)} R(y)$ is zero for $n \geq 2$. To see this, expand $R = \sum_i y_i$; after fixing $y_i = 1$, at least one remaining coordinate is free and cancels in the alternating sum. Thus each parity class carries half of the total relevance mass $n2^{n-1}$, proving normalization. Their average is exactly (22).

For a proper subset $T \subsetneq [n]$, the factor R in each probability cancels the denominator in φ_T. Among vectors with $y_T = 1$, each parity class has $2^{n-|T|-1}$ members, because some coordinate outside T remains free. Division by Z gives the common value in (36). The full monomial is nonzero only at $\mathbf{1}$, which is in the positive parity class; its expectation there is $1/Z$ and is zero in the negative class. ■

Let $\mathcal{G}$ be the n actions of the form $\{i\} \mid ([n] \setminus \{i\})$. Every member has one singleton prefix followed by a b-item block. The next result uses all actions in $\mathcal{A}_{n,b}$, not a two-action restriction chosen to make an example fail.

Theorem 19 (Full-menu parity obstruction) *At reference pricing, the laws in* (35) *have disjoint Bayes action sets, one equal to $\mathcal{G}$ and the other to $\mathcal{A}_{n,b} \setminus \mathcal{G}$. Which law has which set depends on the sign of $(-1)^{b-1}$. For a report consisting of all normalized moments of orders at most b, the exact deterministic two-law minimax regret is*

$$\delta_b = \frac{1}{b(b+1)^2 2^b}, \tag{37}$$

and the exact randomized two-law minimax regret is $\delta_b/2$.

Proof By (16), a degree-n numerator term requires a block of size at least $n - 1 = b$ and a nonempty prefix. With $n = b + 1$ items and cap b, the only such action has a singleton prefix and the remaining items as its block. Thus the full-monomial coefficient is

$$a = \gamma_{1,b} = \frac{(-1)^{b-1}}{b(b+1)}$$

for every action in $\mathcal{G}$, and zero for every other action. All moments of lower order agree at the two laws and at their midpoint ν_n. By Theorem 11, every priced utility at ν_n is κ_n. Its values at p^+, p^- are therefore $\kappa_n + a/(2Z), \kappa_n - a/(2Z)$ for actions in $\mathcal{G}$, and κ_n, κ_n for the other actions. Since $|a|/(2Z) = \delta_b$, this proves the Bayes-set claim.

Any deterministic decoder of the common moment report must choose one of these two action groups. Whichever group it chooses, its regret on one law is δ_b and on the other is zero. This proves both the lower bound and attainability. A randomized decoder putting probability q on one group has regrets $q\delta_b$ and $(1-q)\delta_b$ in some order. Their maximum is minimized at $q = 1/2$ and equals $\delta_b/2$. ■

Corollary 20 (Full-support parity witnesses) *For any $0 < \epsilon < 1$, the laws $p_\epsilon^\pm = (1-\epsilon)\nu_n + \epsilon p^\pm$ have full support, have the same moments through order b, and have the same disjoint Bayes sets as in Theorem 19. Their two-law minimax regrets are $\epsilon\delta_b$ and $\epsilon\delta_b/2$.*

Proof Full support follows from the positive $(1-\epsilon)\nu_n$ component. Moment matching is linear, and every priced utility deviation from κ_n is multiplied by ϵ. The Bayes sets and minimax calculation follow. ■

The obstruction is therefore not confined to zero-probability outcomes. It also identifies the missing order exactly: in the construction, adding the full normalized interaction distinguishes the laws, while every lower-order normalized interaction agrees. The two-law floor is not asserted to be the deficiency of the entire moment fiber; that is a different maximization problem, solved next for three items and uniform fees.

7.2 A Complete Quadratic-Moment Fiber for Three Items

Use the outcome order

$$(100), (010), (110), (001), (101), (011), (111),$$

where the first coordinate is the indicator for item 1. Define

$$p(t) = p_0 + tv, \qquad p_0 = \frac{1}{20}(6,1,4,1,4,1,3), \quad v = (1,1,-2,1,-2,-2,3). \tag{38}$$

Its feasible parameter interval is

$$-\frac{1}{20} \le t \le \frac{1}{40}. \tag{39}$$

Indeed, these are exactly the bounds imposed by coordinatewise nonnegativity; the entries sum to one since $\mathbf{1}^\top v = 0$.

Proposition 21 (The entire fiber) *The laws in* (38) *have the fixed normalized moments*

$$\mu_1 = \frac{11}{20}, \quad \mu_2 = \mu_3 = \frac{9}{40}, \qquad \mu_{12} = \mu_{13} = \frac{3}{20}, \quad \mu_{23} = \frac{3}{40}, \tag{40}$$

while $\mu_{123} = 1/20 + t$. *They are all distributions with the moments* (40)*; there are no other laws in this fiber.*

Proof Direct substitution shows that $\mathbb{E}_{p_0}\varphi_T$ has the stated value and that $\sum_y v_y \varphi_T(y) = 0$ for $1 \le |T| \le 2$. The triple moment is $p(t)(111)/3 = 1/20 + t$. The six quadratic normalized features are independent on the seven nonzero outcomes by Lemma 6. Their matrix has rank six, and their row span contains normalization because $\varphi_1 + \varphi_2 + \varphi_3 = 1$. Its kernel is therefore one-dimensional. The nonzero vector v belongs to that kernel and spans it. Intersecting the affine line through p_0 with the simplex gives exactly (39). ■

The item marginals are fixed as well. For example, the pointwise identity

$$y_i = \varphi_i + \sum_{j \ne i} \varphi_{ij}$$

follows by multiplying φ_i by R. In contrast, the ordinary pair marginals are not fixed: $y_i y_j = 2\varphi_{ij} + \varphi_{123}$ when $n = 3$. The construction is thus not a claim that *all* unnormalized second moments agree.

Table 1: All three-item action risks on the fiber (38).

Action or equal-risk group	Number	$L_{c_\lambda}(A; p(t))$
$1 \mid 2 \mid 3,\ 1 \mid 3 \mid 2$	2	$9/80$
$1 \mid \{2,3\}$	1	$17/240 + \lambda + t/6$
$\{1,2\} \mid 3,\ \{1,3\} \mid 2$	2	$3/40 + \lambda$
$2 \mid 1 \mid 3,\ 3 \mid 1 \mid 2$	2	$11/40$
$2 \mid \{1,3\},\ 3 \mid \{1,2\}$	2	$31/120 + \lambda + t/6$
$\{2,3\} \mid 1$	1	$4/15 + \lambda$
$2 \mid 3 \mid 1,\ 3 \mid 2 \mid 1$	2	$41/120$
$\{1,2,3\}$	1	3λ

Under the pair fee (7), every action's risk is affine in t and λ. Table 1 enumerates the full menu. Grouped actions have identical risks throughout the fiber; braces indicate a resolved block, and vertical bars separate ordered blocks. The last row is present only when $b = 3$; all other rows apply to both $b = 2$ and $b = 3$.

Theorem 22 (Uniform pair fees) *For $n = 3$, $b \in \{2, 3\}$, and*

$$\frac{3}{80} \le \lambda \le \frac{1}{20}, \tag{41}$$

the Bayes risk on the entire fiber is

$$L^*_{c_\lambda}(p(t)) = \min\left\{\frac{9}{80}, \frac{17}{240} + \lambda + \frac{t}{6}\right\}. \tag{42}$$

For every λ strictly inside this interval, no decoder of all normalized moments through degree two is calibrated. Put

$$\alpha = \frac{1}{20} - \lambda, \qquad \beta = \lambda - \frac{3}{80}.$$

The exact deterministic and randomized minimax regrets on this fiber are, respectively,

$$\rho_{\text{det}} = \min(\alpha, \beta), \qquad \rho_{\text{rand}} = \frac{\alpha\beta}{\alpha + \beta} = 80\alpha\beta. \tag{43}$$

An optimal randomized decoder chooses $1 \mid \{2, 3\}$ with probability 80α and a best complete ranking otherwise.

Proof Let F denote either complete action in the first row of Table 1, and let $D = 1 \mid \{2, 3\}$. Every action other than F, D has risk at least that of F on the parameter rectangle (39)–(41). For example, the first-block pair risks are $3/40 + \lambda \ge 9/80$, and the full-block risk is $3\lambda \ge 9/80$. The other complete rankings are strictly worse than F. The two remaining singleton-prefix pair actions have risk at least $31/120 + 3/80 - 1/120 > 9/80$, and $\{2, 3\} \mid 1$ is also worse. This proves (42) for the whole rectangle, not only at sampled points.

The two surviving risks cross at

$$t^*(\lambda) = \frac{1}{4} - 6\lambda.$$

For an interior fee, the crossing lies strictly inside the feasible t interval. At its two endpoints different action groups are uniquely optimal relative to one another, and all other actions are dominated. Since the moment report is constant, the Bayes intersection over the fiber is empty. Theorem 17 gives the failure of calibration.

Choosing F has largest regret α, attained at $t = -1/20$, while choosing D has largest regret β, attained at $t = 1/40$. Other deterministic actions cannot improve on F. This gives the first formula in (43). For randomization, replace any probability on a dominated action by F; this cannot increase risk at any law in the fiber. A mixture putting mass q on D has endpoint regrets $(1-q)\alpha$ and $q\beta$. Within the interval, its regret is convex in t because its risk is affine and Bayes risk is concave. Its maximum is therefore at an endpoint. Balancing the two endpoint regrets gives $q = \alpha/(\alpha+\beta)$ and minimax value $\alpha\beta/(\alpha+\beta)$. Finally, $\alpha + \beta = 1/80$, yielding the displayed forms, including the endpoints where one of the two quantities is zero. ■

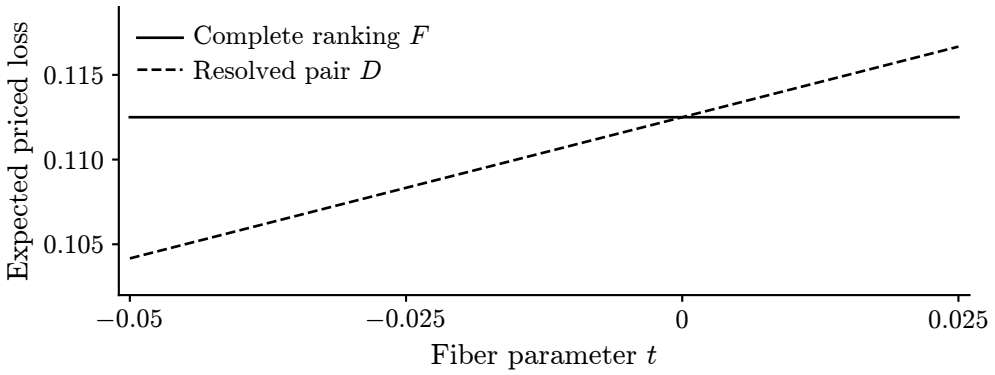


Figure 1: Risks on the entire quadratic-moment fiber at $\lambda = 1/24$. The horizontal curve is the best complete-ranking risk; the increasing curve is the risk of $1 \mid \{2,3\}$. Their lower envelope is the Bayes risk. The common quadratic report cannot distinguish any horizontal location. Decimal tick labels are display conversions of the exact rational endpoints.

At $\lambda = 1/24$, the entire-fiber values are

$$\rho_{\mathrm{det}} = \frac{1}{240}, \qquad \rho_{\mathrm{rand}} = \frac{1}{360}, \qquad q = \frac{2}{3}.$$

Figures 1 and 2 show this information loss as a change in the optimal decision, not as a prediction error on one chosen action. The quantities are absolute AP-loss regret plus fees, not relative performance percentages.

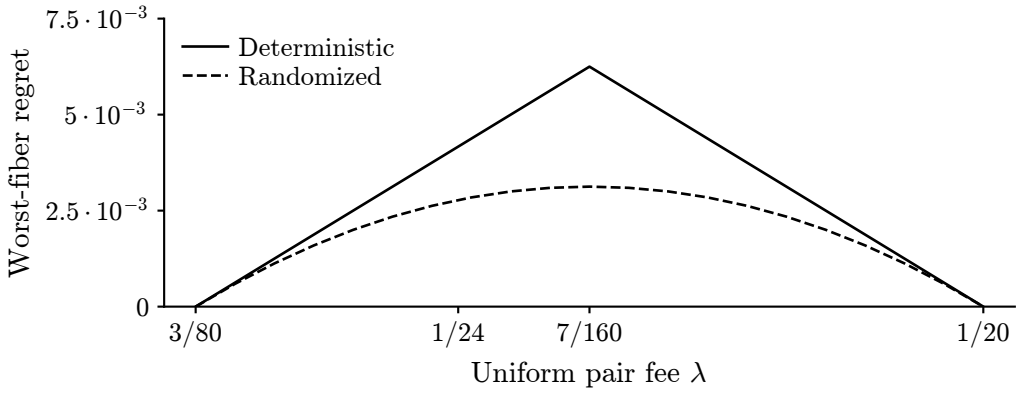


Figure 2: Exact deterministic and randomized deficiencies on the fixed three-item fiber as the uniform pair fee varies. Both are positive in the open interval $(3/80, 1/20)$ and vanish at its endpoints. Randomization reduces, but does not eliminate, the information loss. Data are rational evaluations of Theorem 22; lines connect display samples.

7.3 Explicit Full-Support Laws and Padding with Irrelevant Items

A compact certificate uses two interior points of the same fiber:

$$p(-1/120) = (7/24, 1/24, 13/60, 1/24, 13/60, 1/15, 1/8), \tag{44}$$

$$p(1/120) = (37/120, 7/120, 11/60, 7/120, 11/60, 1/30, 7/40). \tag{45}$$

All seven entries are positive. At fee $1/24$, the first law uniquely selects D with risk $1/9$; the second selects the two F rankings with risk $9/80$. The wrong choice between these groups has regret $1/720$ at either law. Thus the exact two-law deterministic and randomized floors are $1/720$ and $1/1440$. These are smaller than the entire-fiber floors above, because the pair omits the most adverse endpoint distributions. Keeping these two calculations separate is important when using a finite counterexample as evidence.

Corollary 23 (Uniform-fee obstruction for larger menus) *For every $n \geq 3$, every $2 \leq b \leq n$, and every uniform pair fee in $(3/80, 1/20)$, the full collection of normalized moments through degree two is insufficient for calibration on $\mathcal{A}_{n,b}$.*

Proof Embed the three-item fiber by making items $4, \ldots, n$ deterministically irrelevant. Its relevance count and all moments on the first three items are unchanged, while every moment containing an added item is zero.

For any action on all n items, remove the known-irrelevant items from its blocks, discard any empty blocks, and append the removed items as singleton blocks. A relevant item moves weakly earlier in the resulting oracle completion and has the same number of relevant predecessors, so each positive AP summand weakly increases. The pair fee weakly decreases, because removing items from a block cannot increase its number of unordered pairs. The cap is preserved. Thus an optimal decision can always be represented by an action on the

original three items followed by irrelevant singletons. This reduced menu is $\mathcal{A}_{3,2}$ if $b = 2$ and $\mathcal{A}_{3,3}$ if $b \geq 3$.

If a common action were Bayes throughout the embedded fiber, its reduced action would also be Bayes throughout the original fiber: reduction cannot increase risk, and the original action already has Bayes risk. This contradicts Theorem 22. Hence no common action exists, and Theorem 17 applies. ■

The padded laws are not full-support distributions on $\mathcal{Y}_n$; no such claim is needed for this corollary. The reference-priced parity construction provides separate full-support obstructions at higher orders. Finally, insufficiency of a particular quadratic mean map is not itself the exact convex calibration dimension for every uniform fee. The exact lower bound for arbitrary convex surrogates remains the reference-fee result of Theorem 12.

8 Exact Decoding and Finite Validation

The exact subset decoder and rational checks below corroborate the oracle completion, interaction coefficients, contrast ranks, reference-price tie, and full-menu counterexamples. The general results for arbitrary n and b rest on the proofs, not these finite checks.

8.1 Subset Decoding for Additive Fees

Suppose the normalized moments through degree $d = \min(b + 1, n)$ are available. If S is the set already placed and the next nonempty block is $B \subseteq [n] \setminus S$, define

$$V(S, B; \mu) = \sum_{i \in B} \mu_i + \sum_{\emptyset \neq T \subseteq B} \gamma_{|S|,|T|} \left(\sum_{i \in S} \mu_{T \cup \{i\}} - |S| \mu_T \right). \tag{46}$$

For an additive fee $q(S, B)$, set $D(\emptyset) = 0$, initialize all other states to $-\infty$, and update

$$D(S \cup B) = \max\{D(S \cup B), D(S) + V(S, B; \mu) - q(S, B)\}. \tag{47}$$

Theorem 24 (Subset decoding) *For every law p and additive fee, recurrence* (47) *returns Bayes risk $1 - D([n])$; predecessor backtracking returns a Bayes action. Its transition count is*

$$T_{n,b} = \sum_{s=1}^{b} \binom{n}{s} 2^{n-s}. \tag{48}$$

A direct implementation stores $O(2^n + m_d)$ numbers and performs $O(n 2^b T_{n,b})$ arithmetic operations, without a claim about their bit complexity.

Proof Taking expectations in (16) shows that (46) is exactly the utility contributed by B. Earlier blocks matter only through their union S. Hence paths from $\emptyset$ to $[n]$ in the subset graph are in bijection with ordered partitions, and a path weight is expected utility minus total fee. The graph is acyclic, so dynamic programming finds the maximum-weight path. For a block of size s, each subset of the other $n - s$ items is a possible prefix, yielding $\binom{n}{s} 2^{n-s}$ transitions. Equation (46) examines at most $2^b - 1$ subblocks and at most n prefix

Table 2: Exact finite contrast checks. "Actions" is the complete menu at the indicated cap; rank is over the rational utility table.

n	b	Actions	Outcomes	Rank $= D_{n,b}$	Maximum degree
2	1	2	3	1	2
2	2	3	3	2	2
3	1	6	7	4	2
3	2	12	7	6	3
3	3	13	7	6	3
4	1	24	15	8	2
4	2	66	15	13	3
4	3	74	15	14	4
4	4	75	15	14	4
5	1	120	31	13	2
5	2	450	31	24	3
5	3	530	31	29	4
5	4	540	31	30	5
5	5	541	31	30	5

items for each; one value and a predecessor stored as two subset masks per state give the storage bound. The executable position–size decoder validates the moment report once, then probes only the masks required by each transition. ■

The state S is sufficient only for additive fees; a history-dependent fee may require a larger state. The calibrated affine link in Section 6 does not require additivity. Even for $b = 2$, this decoder is exponential in n: it supplies an exact finite decision rule and small-instance oracle, not a scalable retrieval algorithm. This separation between calibrated reporting and structured inference is also central in structured-surrogate analyses such as Osokin et al. (2017).

8.2 Exact Finite Certificates

The artifact uses reduced rational arithmetic and enumerates each ordered set partition once. For every $2 \le n \le 5$ and $1 \le b \le n$, it constructs the complete action–outcome table independently by maximizing the AP definition over all block refinements, evaluating relevance-first completion, and applying the harmonic formula. Boolean Möbius coefficients of the definition table must equal the Newton expansion. Utility-contrast, reference-centered, and constant-augmented ranks must be $D_{n,b}$, $D_{n,b}$, and $D_{n,b} + 1$; the basis, coefficient bound, and all-action reference tie are checked entrywise. At $n = 6$, every complete menu is checked in coefficient space, including the numerator coefficient bound, with exact rational independent-row certificates and 72 definition-level action–outcome comparisons.

The 14 exhaustive cases in Table 2 comprise 71,428 action–outcome entries and 205,074 refinement evaluations; six additional $n = 6$ coefficient cases include menus as large as 4,683

actions. The three-item fiber retains all 13 actions, two full-support witnesses, the rank-six mean matrix, and every affine risk; four rectangle corners certify dominance by affinity. Parity moments and the distinguished top coefficient are checked through $b = 8$ ($n = 9$, 511 outcomes), with complete menus through $b = 5$.

Three negative controls isolate the mechanism: quadratic truncation causes 9,260 mismatches; uniform random block completion leaves all 91 checked numerators quadratic, as Proposition 5 predicts; and a position-independent replacement price destroys the reference tie. The decoder matches exhaustive minimization for three rational laws and four fees in all 14 small cases (168 exact optima); six $b = 2$ runs through $n = 8$ check returned-action risk and (48) without claiming exhaustive optimality.

The current suite defines 47 unit and mutation test methods, including six dedicated full-moment inversion tests for exact round trips, independent inverse oracles, and malformed reports. The remaining methods cover exact-rational input boundaries, relabeling equivariance, monotonicity under block merging, contrast reconstruction, the active-face rank certificate, single-pass report validation, transition-local moment access, constant-size mask predecessors, bibliography consistency, and deterministic-result drift detection. This is the current test inventory; retained resource records keep the counts and timings of their earlier runs. No model is trained. Finite checks detect implementation and algebra errors; the general guarantees rest on Sections 4–7.

9 Discussion

Complete-AP information need not suffice when a system can buy partial resolution. Oracle completion creates a prefix–block interaction: its Newton expansion supplies every contrast direction through degree $b + 1$, and the reference-value price makes those directions simultaneously Bayes-relevant.

Fees determine both which actions matter and, potentially, convex report dimension. At the stated reference schedule the answer is $D_{n,b}$, and one common surrogate for all admissible schedules needs the same dimension. The uniform-pair fiber gives a robust quadratic-report obstruction, not the convex calibration dimension of every uniform fee; free full-block resolution shows why that distinction is necessary.

The model assumes perfect within-block resolution. Resolver error, delay, budgets, and label-acquisition costs would alter the loss and possibly its interaction order. Whole-instance reject formulations optimize a different coverage–risk trade-off (Franc et al., 2023); their strategies are not our block prices or oracle actions.

The exponential decoder implies no restricted-class rate, benchmark gain, or deployment guarantee. Contrast coordinates audit reports; parity/fiber laws test low-order information; the decoder supplies a small-list approximate-inference oracle. These diagnose information deficiency, not resolver or reference-price validity. Computational/statistical efficiency remains separate, and prediction structure must be specified, as in ordinal-regression consistency (Pedregosa et al., 2017).

The outcome space excludes all-zero relevance and otherwise permits arbitrary dependence. Thus the witnesses are exact conditional counterexamples, not prevalence estimates. Within these boundaries, the contrast space, activating fee, squared surrogate, fixed-fiber cri-

terion, and decision-cost obstructions are exact. Code and accompanying artifact materials are available at https://github.com/Haoyi-Zhang/RankCalibrate.

Appendix A. Small Cases and Moment Inversion

This appendix records the smallest geometry and full normalized-moment inversion.

A.1 Two- and Three-Item Geometry

For $n = 2$ in outcome order $(10), (01), (11)$, $U_{1|2} = (1, 1/2, 1)$, $U_{2|1} = (1/2, 1, 1)$, and $U_{\{1,2\}} = (1, 1, 1)$. The full block raises contrast dimension from one to two. At law $(1/4, 1/4, 1/2)$ the losses are $p_{01}/2$, $p_{10}/2$, and $1/8$; their tie has no two-sided simplex direction. At zero fee the full block is universally optimal, showing why rank alone is price-insufficient.

For $n = 3$, $\kappa_3 = 29/36$ and $\nu_3 = (1, 1, 2, 1, 2, 2, 3)/12$. Reference pair fees are $1/12$ first and $1/24$ after a singleton, and the full-block fee is $7/36$; the uniform fiber at $\lambda = 1/24$ instead uses $1/24$ and $1/8$.

A.2 The Feasible Normalized-Moment Polytope

For nonempty $S \subseteq [n]$, write $p_S = p(\mathbf{1}_S)$. All normalized moments invert as

$$p_S = |S| \sum_{T \supseteq S} (-1)^{|T|-|S|} \mu_T \qquad (\emptyset \neq S \subseteq [n]). \tag{49}$$

Indeed, with $w_S = p_S/|S|$ and $\mu_T = \sum_{J \supseteq T} w_J$, the coefficient on w_J is $\sum_{S \subseteq T \subseteq J} (-1)^{|T|-|S|} = \mathbf{1}\{J = S\}$; no independence assumption is used. Hence the full moment polytope is exactly

$$\sum_i \mu_i = 1, \qquad \sum_{T \supseteq S} (-1)^{|T|-|S|} \mu_T \geq 0 \quad (\emptyset \neq S \subseteq [n]). \tag{50}$$

Necessity is $p_S \geq 0$; sufficiency follows from (49). Degree-$d < n$ feasibility is its projection, not a coordinatewise box.

The map $p \mapsto (\mu_T)_{\emptyset \neq T \subseteq [n]}$ is therefore an affine bijection: the Boolean zeta transform of $w_S = p_S/|S|$ is triangular with unit diagonal, and (49) is its inverse. Since $\sum_i \mu_i = \sum_J p_J$, the polytope has affine dimension $2^n - 2$ and its inequalities are the images of the simplex facets $p_S \geq 0$.

For $n = 3$, writing $z = \mu_{123}$ and substituting (40) gives, in Section 7's outcome order,

$$(p_1, p_2, p_{12}, p_3, p_{13}, p_{23}, p_{123}) = \left(\frac{1}{4} + z, z, \frac{3}{10} - 2z, z, \frac{3}{10} - 2z, \frac{3}{20} - 2z, 3z\right).$$

For arbitrary fixed moments through degree two, these seven nonnegativity constraints reduce to the coupled interval

$$\max\{0, \mu_{12} + \mu_{13} - \mu_1, \mu_{12} + \mu_{23} - \mu_2, \mu_{13} + \mu_{23} - \mu_3\} \leq z \leq \min\{\mu_{12}, \mu_{13}, \mu_{23}\}.$$

For (40), $0 \leq z \leq 3/40$. Setting $t = z - 1/20$ gives (38)–(39) and kernel direction $(1, 1, -2, 1, -2, -2, 3)$. Hence this is the whole quadratic-report fiber: its endpoints activate different simplex facets, every interior point has full support, and Theorem 22

changes Bayes action along it. Full moments instead identify the law, so truncation—not normalization—causes the ambiguity.

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